



Artificial Intelligence Capstone Project

Project Title

Smart Resume Analyzer with AI-Based Feedback.

Project Overview

The Smart Resume Analyzer is an AI-powered web application that helps students analyze and improve their resumes based on industry hiring standards. The system evaluates resumes for ATS compatibility, resumes structure, technical skills, and missing keywords, and then provides personalized suggestions for improvement.

This project is designed to give students practical exposure to how Artificial Intelligence is used in recruitment and HR technology platforms.

Project Objective

Develop a web-based application where users can:

- Upload resumes in PDF or DOCX format
 - Extract and process resume content
 - Generate resume scores
 - Check ATS compatibility
 - Analyze role-based skills
 - Receive smart improvement suggestions
 - View analysis results on a dashboard
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Problem Statement

Many students create resumes without understanding:

- ATS (Applicant Tracking System) requirements
- Industry-relevant technical skills
- Resume formatting standards
- Keyword optimization
- Role-specific expectations





This project helps students understand how modern resume screening systems work and how AI can assist in the recruitment process.

Project Difficulty Level

Intermediate Level

Estimated Difficulty: 6/10

This project is designed for college students with basic knowledge of:

- Python
 - Web Development
 - Databases
 - Basic AI/NLP concepts
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Industry Relevance

This project simulates features used in real-world HR-Tech and recruitment platforms.

Industry applications include:

- Resume screening systems
 - ATS software
 - Recruitment automation tools
 - Career guidance platforms
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Students will gain exposure to practical AI implementation in hiring workflows.





Project Modules

Module 1: Resume Upload and Parsing

Description

Users upload resumes in PDF or DOCX format. The system extracts text content from the uploaded resume.

Functionalities

- Upload resume files
- Extract text from documents
- Store resume information

Learning Outcomes

- File handling
- Text extraction
- Backend integration

Module 2: Resume Score Analyzer

Description

The system analyzes the resume based on predefined parameters and generates a resume score.

Analysis Parameters

- Resume structure
- Skills section
- Education details
- Projects section
- Contact information
- Resume completeness





Output

Resume score out of 100.

Learning Outcomes

- Rule-based AI logic
- Data processing
- Resume evaluation systems

Module 3: ATS Keyword Checker

Description

Users select a target job role, and the system compares resume keywords with industry-required keywords.

Example Roles

- Data Analyst
- Web Developer
- AI Engineer
- Cloud Engineer

Example Suggestions

- Add SQL skill
- Include React projects
- Mention cloud technologies

Learning Outcomes

- Keyword extraction
- Basic NLP concepts
- Skill matching systems

Module 4: Smart Feedback System

Description

The system provides improvement suggestions based on resume analysis.





Example Suggestions

- Improve resume formatting
- Add technical projects
- Include certifications
- Add internship experience

Learning Outcomes

- Recommendation logic
- User-focused AI systems
- Decision-making workflows

Module 5: Dashboard and Report Generation

Description

Display all analysis results through a simple and interactive dashboard.

Dashboard Features

- Resume score
- ATS compatibility score
- Missing skills
- Improvement suggestions

Learning Outcomes

- Dashboard creation
- Data visualization
- Frontend-backend integration

Recommended Technology Stack

Frontend

- HTML
- CSS
- Bootstrap or Tailwind CSS
- React.js (Optional)





Backend

- Python Flask

AI/NLP Libraries

- spaCy
- NLTK

Database

- SQLite or MySQL

Deployment

- Render
- Netlify
- Vercel

Project Workflow

1. User uploads resume
2. System extracts resume text
3. Resume data is processed
4. Resume score is calculated
5. ATS keyword matching is performed
6. Suggestions are generated
7. Dashboard displays final analysis

Dataset Suggestions

Students may use:

- Public resume datasets from Kaggle
- Sample job descriptions
- Self-created sample resumes





Expected Deliverables

Students must submit:

- Complete source code
- Project documentation
- PPT presentation
- Project report
- GitHub repository link
- Deployment link
- Demo video

Learning Outcomes

By completing this project, students will learn:

- Practical AI implementation
- Basic Natural Language Processing
- Keyword analysis systems
- Web application development
- Dashboard development
- Deployment process

Scope Limitations

Students are advised NOT to include:

- Deep Learning models
- Large Language Model training
- Complex recommendation engines
- Enterprise-level authentication systems
- Large-scale cloud architecture

The focus should remain on creating a clean, functional, industry-oriented prototype.





Suggested Timeline

Week	Tasks
Week 1	Requirement Analysis and UI Design
Week 2	Resume Upload and Parsing
Week 3	Resume Scoring and ATS Analysis
Week 4	Feedback System and Dashboard
Week 5	Testing, Deployment, and Documentation

Evaluation Criteria

<u>Criteria</u>	<u>Weightage</u>
Functionality	30%
AI Logic Implementation	20%
UI/UX Design	15%
Code Quality	15%
Deployment	10%
Documentation and Presentation	10%

Final Outcome

At the end of the project, students will have developed an AI-powered Resume Analyzer capable of:

- Resume evaluation
- ATS keyword analysis
- Skill gap identification
- Smart feedback generation
- Dashboard reporting

